



# Combining NMR and Computational Approaches to Characterize pH-dependent Switching in Myristoylated Hisactophilin

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## Introduction

- Proteins populate ensembles of conformations on rugged energy landscapes, while function emerges from transitions between states
- Allostery coordinates distant sites of a protein, yet the pathway of signal transmission remains unanswered
- Hisactophilin is an ideal model to study: a histidine-rich, N-myristoylated  $\beta$ -trefoil protein whose pH-dependent membrane switch is driven by a distributed electrostatic-hydrophobic communication network
- Objective:** How do multi-mutations (SAAS, SGGY, GSGGY) and pH alter internal communication networks and myristoyl binding in hisactophilin?

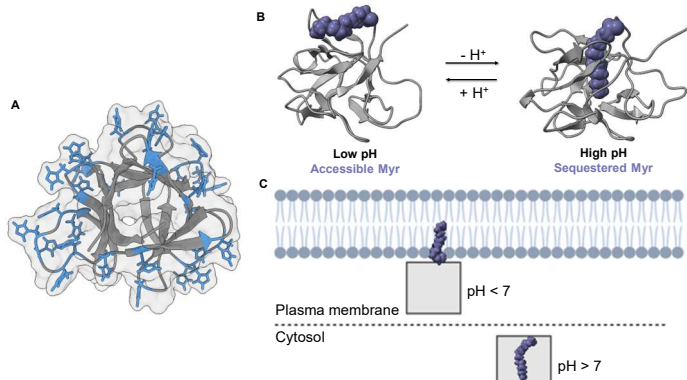
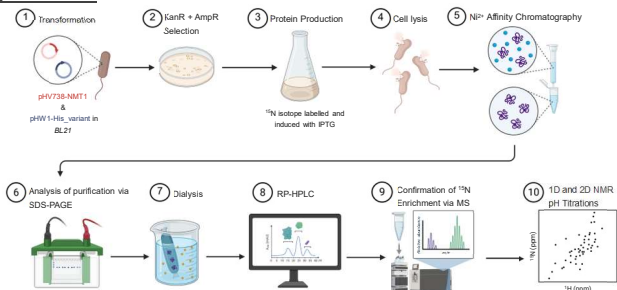


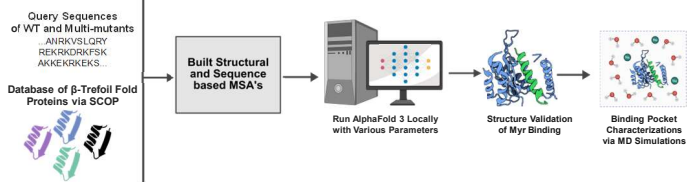
Fig. 1: pH-dependent myristoyl switching of hisactophilin. (A) Crystal structure (PDB: 1HCD) with 31 histidines highlighted in blue. (B) At low pH the myristoyl group (Myr, purple) is solvent-exposed, at high pH it becomes sequestered within the hydrophobic pocket. (C) This switching drives reversible pH-dependent membrane anchoring.

## Methods

### Experimental



### Computational



## Results

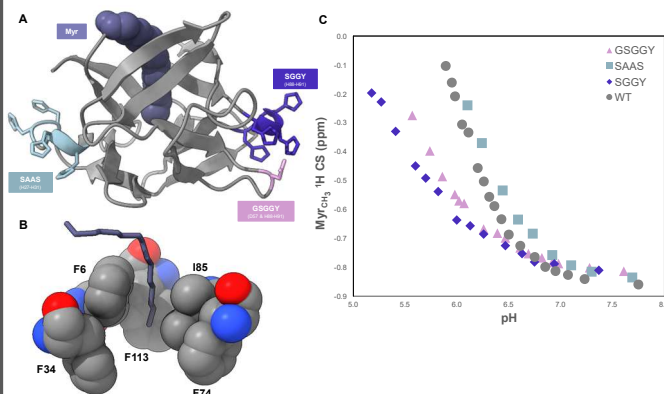


Fig. 2: Hisactophilin variants and pH-dependent myristoyl sequestration. (A) Highlighted residues on WT pertaining to the multi-mutations (SAAS, SGGY, GSGGY) alongside Myr (purple). (B) The hydrophobic pocket formed by F6, F34, F74, I85 and F113 (coloured by heteroatom) that accommodates the Myr (purple) upon sequestration. (C) pH-titration curves tracking the myristoyl  $CH_3$   $^1H$  chemical shift across all variants, showing cooperative sequestration transitions that differ in midpoint and steepness between WT and multi-mutants.

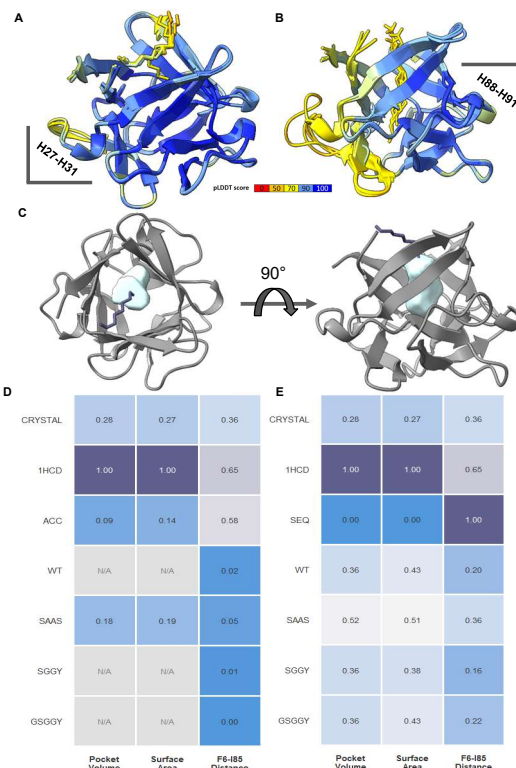


Fig. 3: AlphaFold 3 model confidence and myristoyl pocket analysis of hisactophilin variants. (A-B) Overlay of relaxed AF3 models for WT, SAAS, SGGY, and GSGGY coloured by pLDDT score (blue = high, yellow = low), with Myr in an accessible conformation when covalently linked (A) and sequestered within the hydrophobic pocket when modelled as a ligand (B). (C) Two orthogonal views of the detected myristoyl binding cavity (light blue) in WT; all variants were analyzed equivalently. (D-E) Normalized pocket volume, surface area, and F6-I85 distance for covalently linked (D) and ligand-modelled (E) AF3 structures with references (CRYSTAL, 1HCD, ACC and SEQ); darker shading = more sequestered, N/A = no pocket detected.

## Discussion

- AlphaFold 3 successfully models myristoylated hisactophilin variants, with covalently linked models yielding lower RMSD against reference structures and greater consistency across variants than ligand-modelled counterparts
  - Covalent linkage favours an **accessible state**
  - Ligand modelling favours a **sequestered state**
- SGGY mutant consistently produced the largest structural deviations from reference structures, suggesting this insertion most significantly perturbs the native fold
- Pocket volume and F6-I85 distance are correlated:
  - Larger cavities coincide with greater residue separation, supporting a pocket-opening mechanism that accommodates myristoyl sequestration
- AlphaFold 3 is a powerful tool for capturing and predicting these conformational states, laying the groundwork for future simulations

## Future Directions

- Utilize AlphaFold 3 models as starting states for constant pH molecular dynamic (CpHMD) simulations to capture myristoyl switching between accessible and sequestered conformations across a physiological pH range
- Track real-time binding pocket size changes throughout CpHMD trajectories using GROMACS' trj\_cavity tool, enabling quantitative monitoring of cavity volume as a function of pH and histidine protonation state
- Explore the impact of paired vs. unpaired MSA strategies in AlphaFold 3 to optimize prediction of the myristoyl-bound starting state in hisactophilin

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